

DR. RAJIB BARAN ROY, PhD

CONTACT INFORMATION

Dhaka, Bangladesh

Email: rajib.roy@primeasia.edu.bd | rajibbaran.roy@cqumail.com | r.roy@cqu.edu.au

Google Scholar ID: n2p78lgAAAAJ | Scopus Author ID: 56603588300 | ORCID ID: 0000-0002-1177-0778 | LinkedIn profile identifier: dr-rajib-baran-roy-79a04b259

PROFESSIONAL PROFILE

Electrical Power Systems researcher with 18+ years of academic and research experience. Recently completed a PhD in Electrical Engineering at Central Queensland University, Australia. Research expertise includes power system analysis, smart grids, renewable energy integration, electric ferry and EV charging infrastructure, battery energy storage systems, power quality, harmonic analysis, AI/ML applications, and optimization techniques for modern power systems.

RESEARCH INTERESTS

Power System Analysis; Smart Grids and Microgrids; Renewable Energy Integration; Electric Ferry and EV Charging Infrastructure; Battery Energy Storage Systems; Power Quality and Harmonic Analysis; AI and Machine Learning Applications; Optimization Algorithms.

EDUCATION

PhD in Electrical Engineering, Central Queensland University, Australia (2026)

[Thesis Title: Impacts of Electric Ferry Charging on Power Distribution Network]

MSc in Sustainable Energy Systems, University of Flensburg, Germany (2007)

[Thesis Title: Evaluation of socio-economic impacts of rural electrification through dissemination of photovoltaic technology as compared to conventional electrification strategies in Bangladesh]

BSc in Electrical and Electronic Engineering, BIT, Chittagong (now CUET), Bangladesh (1997)

[Project Title: Design and construction of 5 kW cyclo-converter]

RESEARCH IMPACT

Google Scholar: 620 citations, h-index 13, i10-index 17

Scopus: 381 citations, h-index 9

35+ research outputs

KEY RESEARCH ACHIEVEMENTS

- Australian Government RTP Scholarship recipient.
- Published in high-impact journals including Energies, Energy Reports, IEEE Access and IET Energy Systems Integration.
- Led research on electric ferry charging impacts, BESS integration and distribution network optimization.
- Extensive experience with DIgSILENT PowerFactory, OpenDSS, MATLAB/Simulink and Python.

RESEARCH FUNDING AND COLLABORATIONS

Ministry of Higher Education (MoHE), Malaysia Research Grant

(LRGS/1/2019/UKMUNITEN/6/2)

Green Shore Power Project – Gladstone Port Corporation Limited (GPCL) and CQU

Driving Resilience Project funded by Queensland Reconstruction Authority (QRA)

ACADEMIC AND RESEARCH APPOINTMENTS

Research Engineer, Central Queensland University (2024-2025)

Faculty Member, Primeasia University (2016–Present)

Faculty Member, UITS (2009–2016)

Faculty Member, UITS (2004–2006)

Faculty Member, IUBAT (1998–2004)

TECHNICAL SKILLS

MATLAB, Python, OpenDSS, DigSILENT PowerFactory, PSCAD, HOMER Pro, RETScreen, PowerWorld Simulator

JOURNAL PAPERS

[1] Rajib Baran Roy, Sanath Alahakoon and Piet Janse Van Rensburg, “Optimized Electric Ferry Charging Impacts on Shoreside Electricity Network”, International Journal of Energy Research, Willy Publications, 2026 (accepted).

[2] Rajib Baran Roy, Sanath Alahakoon and Piet Janse Van Rensburg, “Impact of Coordinated Electric Ferry Charging on Distribution Network Using Metaheuristic Optimization”, Energies, vol. 18, no. 11, pp. 2805, 2025. DOI: 10.3390/en18112805

[3] Rajib Baran Roy, Sanath Alahakoon, Piet Janse Van Rensburg and Shantha Jayasinghe Arachchillage, “Impact analysis of uncoordinated electric ferry charging on distribution network”, e-Prime - Advances in Electrical Engineering, Electronics and Energy, vol. 10, pp. 100783, 2024. DOI: 10.1016/j.prime.2024.100783

[4] Rajib Baran Roy, Sanath Alahakoon, Piet Janse Van Rensburg and Shantha Jayasinghe Arachchillage, “Impact analysis on distribution network due to coordinated electric ferry charging”, IET Energy Systems Integration, vol. 6, no. 4, pp. 638–663, 2024. DOI: 10.1049/esi2.12165

[5] Rajib Baran Roy, Sanath Alahakoon, Shantha Jayasinghe Arachchillage and Saifur Rahman, “Optimizing dynamic electric ferry loads with intelligent power management”, Energy Reports, vol. 9, pp. 5952-5963, 2023. DOI: 10.1016/j.egyr.2023.05.029

[6] Sanath Alahakoon and Rajib Baran Roy, “Optimizing Load Frequency Control in Standalone Marine Microgrids Using Meta-Heuristic Techniques”, Energies, vol. 16, no. 13, pp. 4846, 2023. DOI: 10.3390/en16134846

[7] Ferdous Al Hossain, Md. Rokonzaman, Nowshad Amin, Jianmin Zhang, Mahmuda Khatun Mishu, Wen-Shan Tan, Md. Rabiul Islam and Rajib Baran Roy, “Probabilistic Load Flow–Based Optimal Placement and Sizing of Distributed Generators”, Energies, vol. 14, no. 23, pp. 7857, 2021. DOI: 10.3390/en14237857

[8] Md. Rokonzaman, Mahmuda Khatun Mishu, Nowshad Amin, Mithulananthan Nadarajah, Rajib Baran Roy, Kazi Sajedur Rahman, Adamu Muhammad Buhari, Shuza Binzaid, Mohammad Shakeri and Jagadeesh Pasupuleti, “Self-Sustained Autonomous Wireless Sensor Network with Integrated Solar Photovoltaic System for Internet of Smart Home-Building (IoSHB) Applications”, Micromachines, vol. 12, no. 6, pp. 653, 2021. DOI: 10.3390/mi12060653

[9] Rajib Baran Roy, Sanath Alahakoon and Shantha Jayasinghe Arachchillage, “Grid Impacts of Uncoordinated Fast Charging of Electric Ferry”, Batteries, vol. 7, no. 1, pp. 13, 2021. DOI: 10.3390/batteries7010013

[10] Rajib Baran Roy, Md. Rokonzaman, Nowshad Amin, Mahmuda Khatun Mishu, Sanath Alahakoon, Saifur Rahman, Nadarajah Mithulananthan, Kazi Sajedur Rahman, Mohammad Shakeri and Jagadeesh Pasupuleti, “A comparative performance analysis of ANN algorithms for

MPPT energy harvesting in solar PV system", IEEE Access, vol. 9, pp. 102137-102152, 2021. DOI: 10.1109/access.2021.3096864

[11] Md. Ruhul Amin, Rajib Baran Roy and Md. Mahmudul Hasan, "Modeling and Optimization of Decentralized Microgrid System for St. Martin's Island in Bangladesh", International Journal of Energy, Information and Communications, vol. 5, issue. 5, pp. 1-12, 2014. DOI: 10.14257/ijeic.2014.5.5.01

[12] Md. Ruhul Amin and Rajib Baran Roy, "Determination of Electric Vehicle Power Consumption for Assorted Pavement Gradient Using Matlab/Simulink", TELKOMNIKA Indonesian Journal of Electrical Engineering (IJECS), vol. 12, no. 9, pp. 6602-6611, 2014. DOI: 10.11591/ijeecs.v12.i9.pp6602-6611

[13] Rajib Baran Roy and Md. Rokonzaman, "STATCOM Model for Integration of Wind Turbine to Grid", TELKOMNIKA Indonesian Journal of Electrical Engineering (IJECS), vol. 12, no. 9, pp. 6519-6525, 2014. DOI: 10.11591/ijeecs.v12.i9.pp6519-6525

[14] Md. Ruhul Amin, Md. Mahmudul Hasan and Rajib Baran Roy, "Roadmap to Smart Grid Technology: A Review of Smart Information and Communication System", International Journal of Control and Automation (IJCA), vol. 7, no. 8, pp. 407-418, 2014. DOI: 10.14257/ijca.2014.7.8.37

[15] Md. Ruhul Amin and Rajib Baran Roy, "Determination of Volume of Capacitor Bank for Static VAR", International Journal of Electrical and Computer Engineering (IJECE), vol. 4, no. 4, pp.5803, 2014. DOI: 10.11591/ijece.v4i4.5803

[16] Rajib Baran Roy and Md. Ruhul Amin, "A Paper of Determination of Controlling Characteristics of the Monopolar HVDC system", International Journal of Hybrid Information Technology, vol. 7, no. 3, pp. 105-120, 2014. DOI: 10.14257/ijhit.2014.7.3.12

[17] Md. Jahangir Alam, Guo Qing Hu, Md. Rabiul Alam, Rajib Baran Roy and Md. Golam Kibria, "Improved ICT Grid Model for Development of the Bangladesh", Applied Mechanics and Materials, vol. 411-414, pp. 2200-2207, 2013. DOI: 10.4028/www.scientific.net/amm.411-414.2200

[18] Rajib Baran Roy and Rajib Baran Roy, "A comprehensive study of potential zone and low head hydro system in Bangladesh", International Journal of Innovative Technology and Creative Engineering (IJITCE), vol. 3, no. 6, pp. 91-96, 2013.

[19] Rajib Baran Roy and Md. Ruhul Amin, "Design and Construction of Single Phase Cycloconverter," International Journal of Recent Technology and Engineering (IJRTE), vol. 1, issue 3, pp. 75-82, 2012.

CONFERENCE PAPERS

[1] Rajib Baran Roy, Sanath Alahakoon and Piet Janse Van Rensburg, "Coordinated Electric Ferry Charging Impacts by Balanced Hybrid Metaheuristic Optimization", 2025 IEEE PES GTD Grand International Conference and Exposition Asia (GTD Asia), pp. 337-342. DOI: 10.1109/GTDAsia60461.2025.11313286

[2] Rajib Baran Roy, Sanath Alahakoon and Piet Janse Van Rensburg, "Impacts of Uncoordinated Electric Ferry Charging on Distribution Network", 2024 International Conference on Sustainable Technology and Engineering (i-COSTE), IEEE, pp. 1-6. DOI: 10.1109/i-COSTE63786.2024.11024706

[3] Rajib Baran Roy, Sanath Alahakoon, Piet Janse Van Rensburg and Shantha Jayasinghe Arachchilage, "Optimal Capacities of Electric Ferry Charging Stations by Hybrid Metaheuristic Algorithm", 2024 IEEE 34th Australasian Universities Power Engineering Conference (AUPEC), pp. 1-6, 2024. DOI: 10.1109/AUPEC62273.2024.10807529

[4] Rajib Baran Roy, Sanath Alahakoon, Piet Janse Van Rensburg and Shantha Jayasinghe Arachchilage, "Analysis of Electric Ferry's Coordinated Charging Impacts on Distribution

Network", 2024 7th International Conference on Renewable Energy and Power Engineering (REPE), pp. 74-79, 2024. DOI: 10.1109/REPE62578.2024.10809422

[5] Rajib Baran Roy, Sanath Alahakoon, Piet Janse Van Rensburg, Enamul Basher and Raid Mollik Babu, "Impacts on Distribution Network due to Coordinated Electric Ferry Charging", 2024 7th International Conference on Renewable Energy and Power Engineering (REPE), IEEE, pp. 1-6. DOI: 10.1109/ICISSET62123.2024.10939967

[6] Rajib Baran Roy, Sanath Alahakoon, Umme Mumtahina and Ashfaque Ahmed Chowdhury, "PSO Tuned Marine Microgrid Load Frequency Control", 23rd International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME), pp. 1-6, IEEE, 2022. DOI: 10.1109/ICECCME57830.2023.10252104

[7] Rajib Baran Roy and Sanath Alahakoon, "Peak Shaving Approach of Distribution Network", 2022 International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON), pp.1-6, IEEE. DOI: 10.1109/SMARTGENCON56628.2022.10083737

[8] Rajib Baran Roy, Sanath Alahakoon Rensburg and Shantha Jayasinghe Arachchillage, "Microgrid and Fleet to Grid Operation of a Hybrid Electric Ferry", 2021 31st Australasian Universities Power Engineering Conference (AUPEC) pp. 1-6, IEEE. DOI: 10.1109/AUPEC52110.2021.9597730

[9] Rajib Baran Roy, Jerome Cros, Anup Nandi and Towsif Ahmed, "Maximum Power Tracking by Neural Network", 2020 8th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO), pp. 89-93, IEEE. DOI: 10.1109/ICRITO48877.2020.9197882

[10] Rajib Baran Roy, Jerome Cros, Enamul Basher and Shaiyek Md Buland Taslim, "Hybrid Power System Model for Off-Grid Rural Areas of Bangladesh", 2017 IEEE Region 10 Humanitarian Technology Conference (R10-HTC), pp. 489-493. DOI: 10.1109/R10-HTC.2017.8289005

[11] Rajib Baran Roy, Jerome Cros, Enamul Basher and Shahana Akhter, "Power compensation by DSTATCOM plus SCESS", 2017 4th International Conference on Advances in Electrical Engineering (ICAEE), pp. 32-37, IEEE. DOI: 10.1109/ICAEE.2017.8255322

[12] Rajib Baran Roy, Md Rokonzaman and Mohammed Hossam-E-Haider, "Design and analysis of the power electronic transformer for power quality improvement", 2015 International Conference on Electrical Engineering and Information Communication Technology (ICEEICT), pp. 1-5, IEEE. DOI: 10.1109/ICEEICT.2015.7307478

[13] Rajib Baran Roy, Enamul Basher, Joshua Mithil Biswas, "Design and analysis of isolated integrated charger for PHEV", 8th International Conference on Electrical and Computer Engineering, pp. 473-476, IEEE. DOI: 10.1109/ICECE.2014.7026991

[14] Md Ruhul Amin and Rajib Baran Roy, "Design and simulation of wireless stationary charging system for hybrid electric vehicle using inductive power pad in parking garage", The 8th International Conference on Software, Knowledge, Information Management and Applications (SKIMA 2014), pp. 1-5, IEEE. DOI: 10.1109/SKIMA.2014.7083516

[15] Rajib Baran Roy, Enamul Basher, Rojoba Yasmin and Md Rokonzaman, "Fuzzy logic based MPPT approach in a grid connected photovoltaic system", The 8th International Conference on Software, Knowledge, Information Management and Applications (SKIMA 2014), pp. 1-6, IEEE. DOI: 10.1109/SKIMA.2014.7083525

[16] Rajib Baran Roy and Md Ruhul Amin, "Analysis of Proposed Controlling Topology of the HVDC link between Bangladesh and India", 2014 9th International Forum on Strategic Technology (IFOST), pp. 390-395, IEEE. DOI: 10.1109/IFOST.2014.6991147

[17] Md. Ruhul Amin and Rajib Baran Roy, "Design of microcontroller-based Thyristor Controlled three-phase static volt-ampere reactive compensator", 2014 International Conference on Informatics, Electronics & Vision (ICIEV), pp. 1-6, IEEE, DOI: 10.1109/ICIEV.2014.6850694

BOOK PUBLICATION

[1] Md. Ruhul Amin and Rajib Baran Roy, "Electrical Power System Analysis: Control of Voltage and Reactive Power", LAP LAMBERT Academic Publishing, ISBN: 978-3-659-39552-9, 2013.

RESEARCH SUPERVISION

Supervised 60+ undergraduate projects and multiple postgraduate research students.

TEACHING EXPERIENCE

Power Systems, Renewable Energy, Power Electronics, Electrical Machines, Smart Grid and related courses.

SCHOLARSHIPS AND AWARDS

Australian Government RTP Scholarship for PhD Studies
DAAD Scholarship for Master's Degree
Best Conference Paper Award (SKIMA 2014)

PROFESSIONAL MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE)
Institution of Engineers Bangladesh (IEB)

PROFESSIONAL TRAINING

HOMER Pro and HOMER Grid; PLC Applications, Microcontroller-Based Design; Renewable Hydrogen; Operation and Maintenance of Power Station and Substation; Integration of Green Electricity to Grid Network-based field studies in the Shetland Islands, UK

REFERENCES

Dr. Sanath Alahakoon, Associate Professor, Department of Engineering and Technology, Central Queensland University, Queensland, Australia. Email: s.alahakoon@cqu.edu.au

Dr. Piet Janse Van Rensburg, Lecturer, Department of Engineering and Technology, Central Queensland University, Queensland, Australia. Email: p.jansevanrensburg@cqu.edu.au

Dr. Shantha Jayasinghe Arachchillage, Engineer, AEMO (Australian Energy Market Operator), Melbourne, Victoria, Australia. Email: gamini.jayasinghearachchillage@aemo.com.au

Dipl.-Ing. Wulf Boie, Professor, Department of Energy and Environmental Management, University of Flensburg, Munketoft 3b, Flensburg, 24937, Germany. E-mail: boie@uni-flensburg.de

Dr. Olav Hohmeyer, Professor, Department for Sustainable Energy Transition, University of Flensburg, Munketoft 3b, Flensburg, 24937, Germany. E-mail: hohmeyer@uni-flensburg.de

Dr. Sampad Ghos, Professor, Department of Electrical and Electronic Engineering, Chittagong University of Engineering and Technology, Chittagong, Bangladesh. Contact No: 880-1621277369, Email: sampad@cuet.ac.bd